



THE SCOTS COLLEGE

Year 12 Extension II Mathematics ASSESSMENT TASK 1

Friday 17th February 2012
Format: 45 minute examination

Weighting of Task: 10%

Instructions:

- ◆ **Do not open this paper until instructed to do so.**
- ◆ Show ALL necessary working to gain full marks.
- ◆ Board Approved calculators may be used.
- ◆ Diagrams are NOT to scale.
- ◆ Marks may be deducted for careless or incomplete working.
- ◆ **Answer the free response questions (Section A) on the multiple choice answer sheet provided.**
- ◆ **Answer Sections B and C on the separate answer sheets provided.**
- ◆ **Start each question on a new page.**
- ◆ Attempt all questions

Section	Topics	Marks
A	Free Response questions	/5
B	Complex Numbers	/20
	Graphs	/15
	TOTAL	/40

Section A – MULTIPLE CHOICE QUESTIONS (5 Marks)

Each question is worth **1 mark**.

Circle the correct response on the answer sheet paper.

1. In $x + iy$ form, $(2 - 3i)^2$ is equivalent to:

(A) $-5 - 12i$

(B) $-5 + 12i$

(C) $13 - 12i$

(D) $13 + 12i$

2. The graph of the function with rule $f(x) = 2x - \frac{1}{x^2}$ has:

(A) one asymptote and a local minimum at $(-1, -3)$

(B) one asymptote and a local maximum at $(-1, -3)$

(C) two asymptotes and a local minimum at $(-1, -3)$

(D) two asymptotes and a local maximum at $(-1, -3)$

3. The distance between the two points z and $-\bar{z}$ in the complex plane is given by:

(A) $2 |\operatorname{Re}(z)|$

(B) $2 |\operatorname{Im}(z)|$

(C) $2|z|$

(D) $2 |\operatorname{Re}(z)| + 2 |\operatorname{Im}(z)|$

4. The slope of the curve $2x^3 - y^2 = 7$ at the point $(2,3)$ is given by:

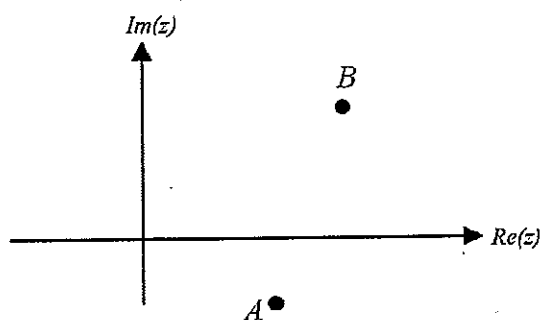
(A) -4

(B) -2

(C) 2

(D) 4

5.



The diagram above shows the points A and B representing the complex numbers $4-2i$ and $6+4i$ respectively. The locus of the circle with diameter AB is given by the equation $|z-k|=r$, where k is a complex number and r is a positive real number. Which of the following statements is true?

(A) $k=-5-i$ and $r=\sqrt{10}$

(B) $k=5+i$ and $r=\sqrt{10}$

(C) $k=-5-i$ and $r=10$

(E) $k=5+i$ and $r=10$

END OF SECTION A

Section B - EXTENDED RESPONSES (35 Marks)

- Show ALL necessary working to gain full marks.
- *Write your responses on writing paper supplied.*
- Clearly write your name and class teacher on each page to be marked.
- Begin each question on a new page.

Section B - QUESTION 1: Complex Numbers (20 Marks)

- (a) If $z = (1 + \sqrt{3}i)(1 + i)$,
- (i) Express z in the form $x + iy$ where x and y are real numbers 2
- (ii) Express both $(1 + \sqrt{3}i)$ and $(1 + i)$ in mod-arg form and then show that $z = \sqrt{8} \operatorname{cis}\left(\frac{7\pi}{12}\right)$ 2
- (iii) Hence find the exact values of $\cos \frac{5\pi}{12}$ and $\sin \frac{5\pi}{12}$ 2
- (iv) What is the smallest positive integer n for which z^n is real 1
- (b) Let $z = \operatorname{cis} \frac{\pi}{4}$ 3

On the **same** Argand plane, plot **and** label carefully **all** the points:

$$z, \bar{z}, -2z, iz, z^2, 2z$$

- (c) (i) Express $-2 - 2\sqrt{3}i$ in modulus-argument form. 2
- (ii) Evaluate $(-2 - 2\sqrt{3}i)^6$ using De Moivre's theorem. 2

Section B - QUESTION 1: (continued)

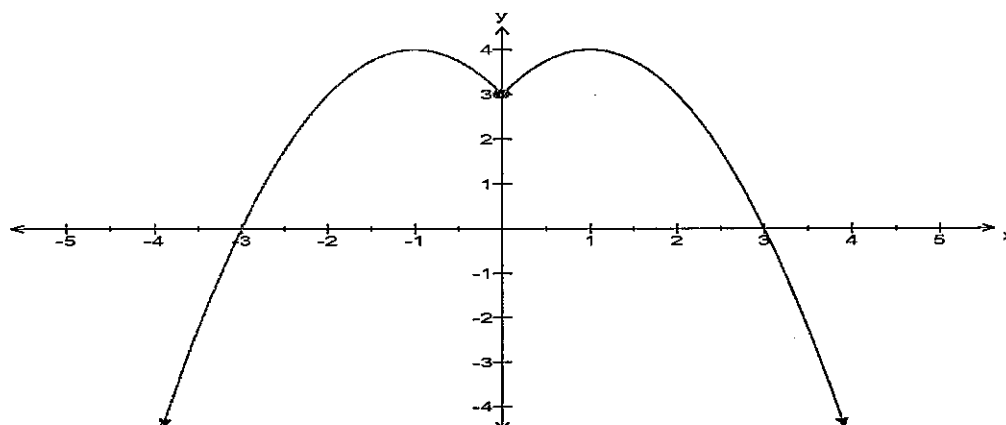
- (d) If ω is a complex root of unity and using the fact that $1 + \omega + \omega^2 = 0$, 3
evaluate $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$

- (e) On an Argand diagram, neatly shade the region that holds 3
simultaneously for both inequalities:

$$|z - (2 + i)| \leq \sqrt{5} \text{ and } \text{Arg } z < \frac{\pi}{12}$$

Section B - QUESTION 2: Graphs (15 Marks) Start a new answer page

- (a) The sketch below is of the even function $y = f(x)$.



On separate number planes sketch each of the following, clearly showing all important features.

- | | | |
|-------|----------------------|---|
| (i) | $y = f(x) - 2$ | 1 |
| (ii) | $y = f(x - 2)$ | 1 |
| (iii) | $y = f(x) $ | 1 |
| (iv) | $y^2 = f(x)$ | 2 |
| (v) | $y = \frac{1}{f(x)}$ | 2 |

Section B - QUESTION 2: Graphs (continued)

- (b) Consider the function $f(x) = \frac{e^x - 1}{e^x + 1}$
- (i) Show that the function is odd 1
 - (ii) Show that the function is always increasing 2
 - (iii) Find $f'(0)$ 2
 - (iv) Sketch $y = f(x)$ showing any asymptotes 2
 - (v) Use your graph to evaluate a set of values of M 1
such that $\frac{e^x - 1}{e^x + 1} = Mx$ has 3 real solutions.
-

END OF SECTION B
END OF EXAMINATION

Yr 12 Extension 2
Task 1 2012
Solutions.

Multi-choice

- | | | |
|------|------|------|
| 1. A | 3. A | 5. B |
| 2. D | 4. D | |

Section B

1. a) (i) $z = (1 + \sqrt{3}i)(1 + i)$
 $= 1 + i + \sqrt{3}i - \sqrt{3}$ ✓
 $= (1 - \sqrt{3}) + i(1 + \sqrt{3})$ ✓

(ii) $1 + \sqrt{3}i = 2 \operatorname{cis} \frac{\pi}{3}$ ✓
 $1 + i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$

$$\begin{aligned} (2 \operatorname{cis} \frac{\pi}{3})(\sqrt{2} \operatorname{cis} \frac{\pi}{4}) &= 2\sqrt{2} \operatorname{cis} \left(\frac{\pi}{3} + \frac{\pi}{4}\right) \checkmark \\ &= 2\sqrt{2} \operatorname{cis} \left(\frac{7\pi}{12}\right) \\ &= \sqrt{8} \operatorname{cis} \left(\frac{7\pi}{12}\right) \end{aligned}$$

(iii) $\sqrt{8} \cos \left(\frac{7\pi}{12}\right) = 1 - \sqrt{3}$ and $\sqrt{8} \sin \left(\frac{7\pi}{12}\right) = 1 + \sqrt{3}$.

Equating real/imaginary parts.

$$\therefore \cos \left(\frac{7\pi}{12}\right) = \frac{1 - \sqrt{3}}{\sqrt{8}} \quad \sin \left(\frac{7\pi}{12}\right) = \frac{1 + \sqrt{3}}{\sqrt{8}}$$

$$\begin{aligned} \cos \left(\frac{5\pi}{12}\right) &= \cos \left(\pi - \frac{7\pi}{12}\right) \\ &= \cos \pi \cos \frac{7\pi}{12} + \sin \pi \sin \frac{7\pi}{12} \\ &= -\left(\frac{1 - \sqrt{3}}{\sqrt{8}}\right) \checkmark \\ \text{or } \frac{\sqrt{3} - 1}{\sqrt{8}} \end{aligned}$$

(12.)

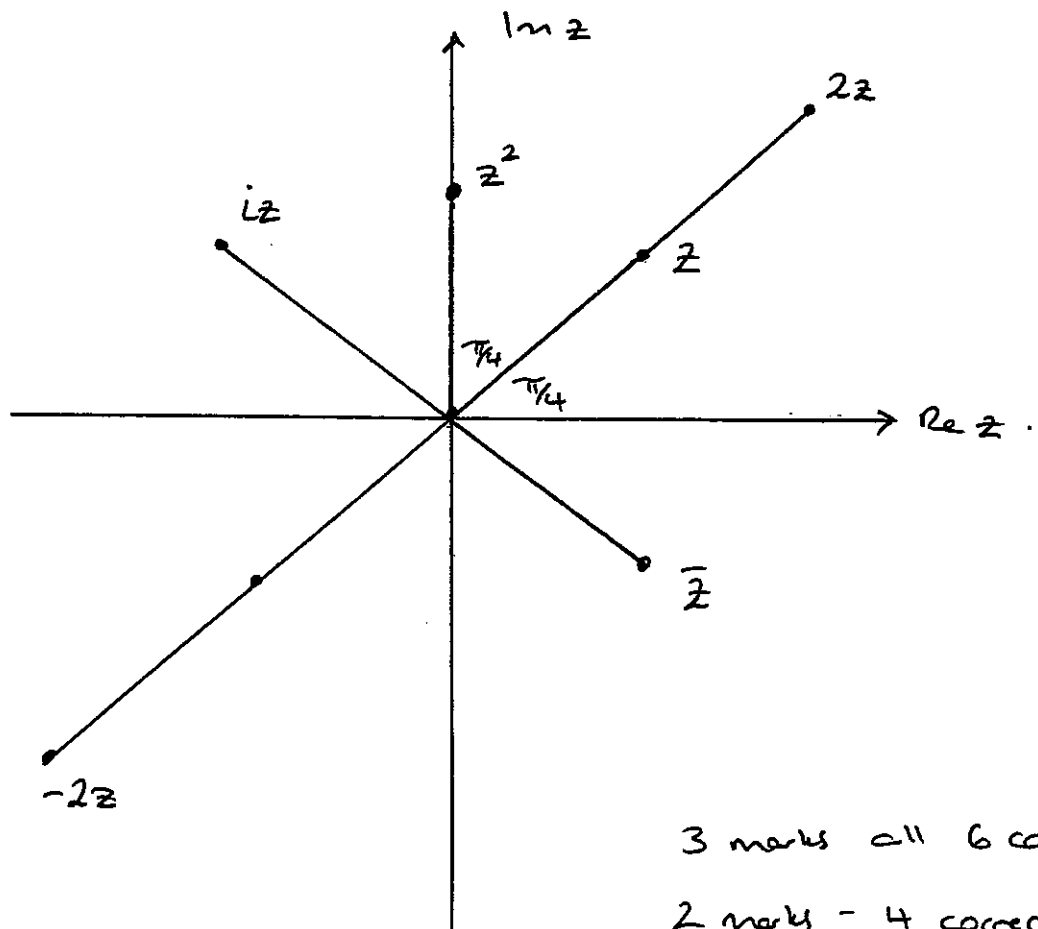
$$\begin{aligned}
 \sin\left(\frac{5\pi}{12}\right) &= \sin\left(\pi - \frac{7\pi}{12}\right) \\
 &= \sin \pi \cos \frac{7\pi}{12} + \cos \pi \cdot \sin \frac{7\pi}{12} \\
 &= + \sin\left(\frac{7\pi}{12}\right) \\
 &= + \left(\frac{1+\sqrt{3}}{\sqrt{8}}\right) \quad \checkmark
 \end{aligned}$$

(iv) Smallest integer of n :

$$z^n = \left[\sqrt{8} \operatorname{cis}\left(\frac{7\pi}{12}\right) \right]^n$$

In order for z^n to be real $n=12$. $\checkmark$

b)



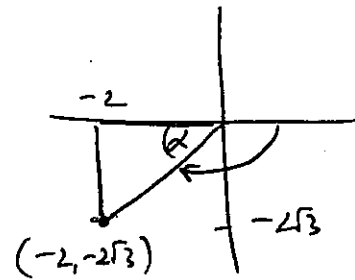
3 marks all 6 correct

2 marks - 4 correct

1 mark - 2 correct

$$c) (i) -2 - 2\sqrt{3}i$$

$$= 4 \operatorname{Cis}\left(-\frac{2\pi}{3}\right)$$



$$\text{modulus} = \sqrt{(-2)^2 + (-2\sqrt{3})^2}$$

$$= 4 \quad \checkmark$$

$$\text{Arg} = \pi - \tan^{-1}\left(\frac{b}{a}\right)$$

$$= -\frac{2\pi}{3} \quad \checkmark$$

$$(ii) (-2 - 2\sqrt{3}i)^6 = \left[4 \operatorname{Cis}\left(-\frac{2\pi}{3}\right)\right]^6$$

$$= 4^6 \operatorname{Cis}(-4\pi) \quad \checkmark$$

$$= 4^6 \text{ (or 4096)} \quad \checkmark$$

$$d) \text{ using } 1 + \omega + \omega^2 = 0$$

$$(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2) = (-3\omega - \omega)(-\omega^2 - 8) \quad \checkmark$$

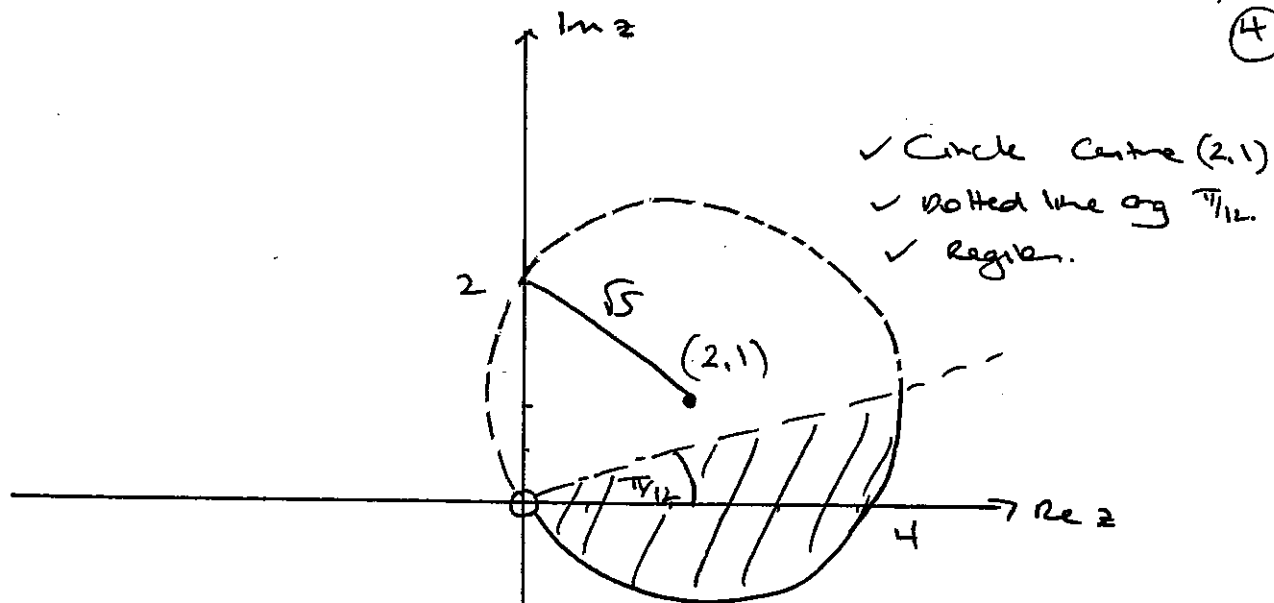
$$= (-4\omega)(-\omega^2 - 8)$$

$$= 36\omega^3 \quad \checkmark$$

$$= 36 \quad \checkmark$$

e)

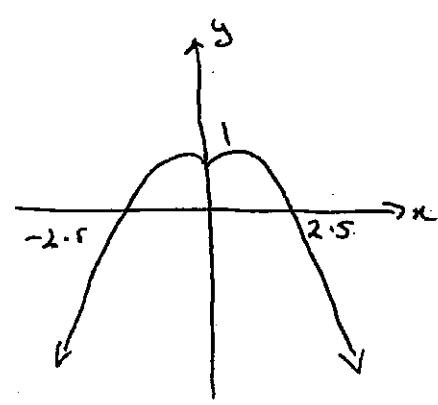
(4)



Note :
 - 1 mark if no open circle at $(0, 0)$.

Section B

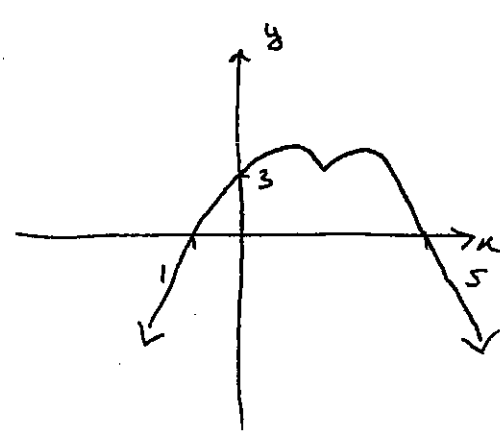
Q 2/ a) (i)



$$y = f(x) - 2$$

✓ shape + y intercept.

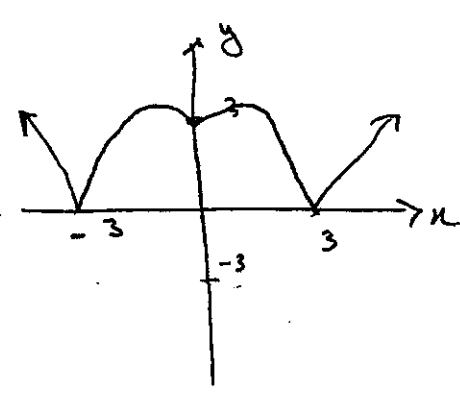
(ii)



$$y = f(x-2)$$

✓ shape + x intercepts / y-int.

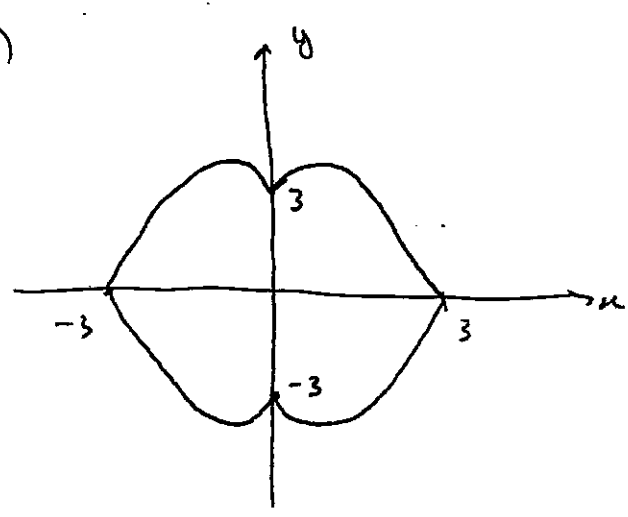
(iii)



$$y = |f(x)|$$

✓ shape

(iv)



$$y^2 = f(x)$$

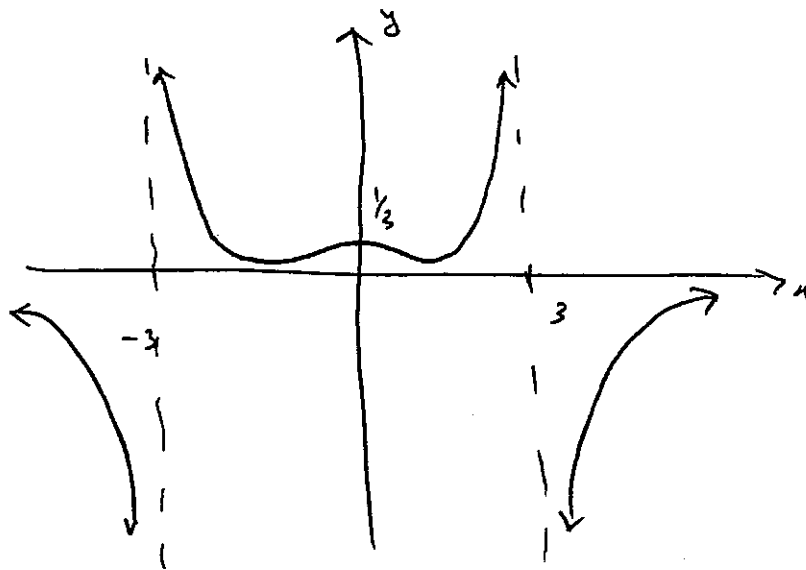
✓ shape

✓ x/y int.

2a) (v)

$$y = \frac{1}{f(x)}$$

(6)



Shape ✓

y-int ✓

(7)

$$\therefore \text{Q2(b)(i)} \quad f(x) = \frac{e^x - 1}{e^x + 1}$$

$$f(x) = \frac{e^{-x} - 1}{e^{-x} + 1}$$

$$= \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1}$$

$$= \frac{1 - e^x}{1 + e^x}$$

$$-f(x) = -\left(\frac{e^x - 1}{e^x + 1}\right)$$

$$= \frac{1 - e^x}{1 + e^x} \quad \checkmark$$

$$f(x) = -f(x) \quad \text{hence odd.}$$

$$(ii) \quad f'(x) = \frac{(e^x + 1) \cdot e^x - (e^x - 1) e^x}{(e^x + 1)^2}$$

$$= \frac{e^{2x} + e^x - e^{2x} + e^x}{(e^x + 1)^2} \quad \checkmark$$

$$= \frac{2e^x}{(e^x + 1)^2} \quad \checkmark$$

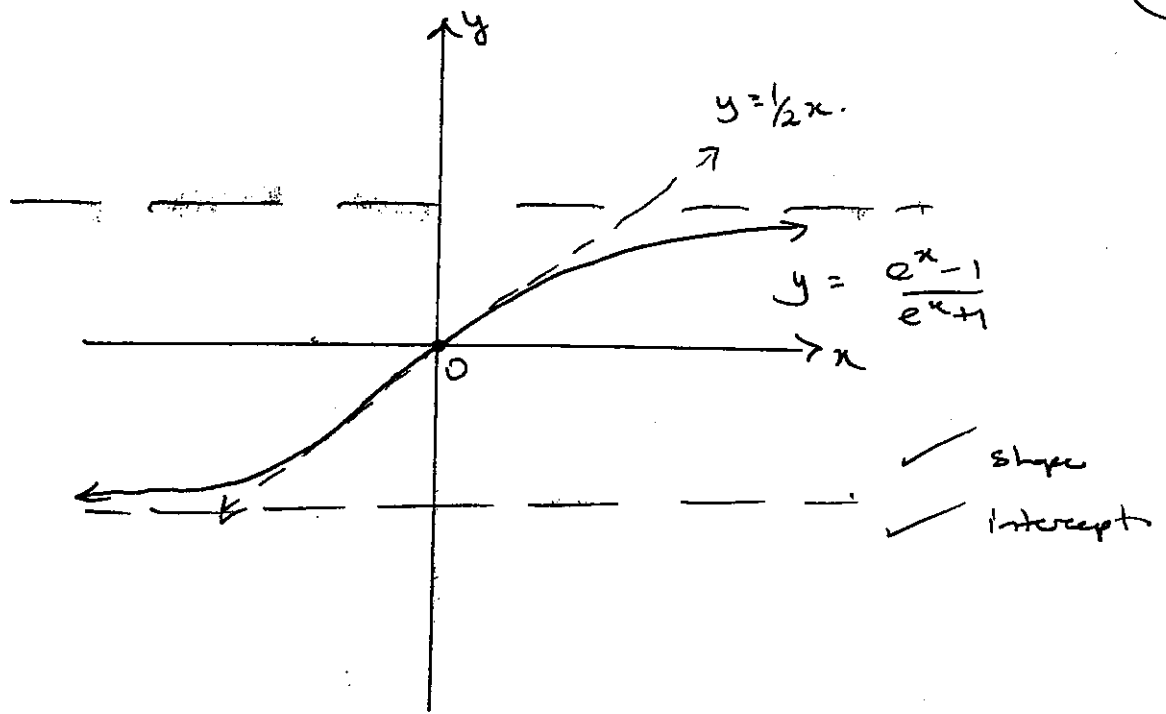
$f'(x) > 0$ for all values of x .

Hence monotonically increasing.

$$(iii) \quad f'(0) = \frac{2e^0}{(e^0 + 1)^2} \quad \checkmark$$

$$= \frac{1}{2} \quad \checkmark$$

(iv) Over pg



$$\text{as } x \rightarrow \infty, \frac{e^x - 1}{e^x + 1} \rightarrow 1$$

$$\text{as } x \rightarrow -\infty, \frac{e^x - 1}{e^x + 1} \rightarrow -1 \quad \left[\text{i.e. } \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1} \right]$$

both $\rightarrow 0$

(iv) $\frac{e^x - 1}{e^x + 1} = Mx$ has 3 solutions (real)
 then line $y = Mx$ intersects $f(x)$ at
 3 points.

line must have gradient between
 $M = 0$ and $M = \frac{1}{2}$.

$$0 < M < \frac{1}{2}$$

